STA 301

CURRENT SOLVED PAPERS

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COMPILED BY

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Current Solved Paper 2021

STA 301

Q No1 Find $\hat{P}_c$ by using information:

We are given that

$\hat{P}_1 = 0.31$, $\hat{P}_2 = 0.24$

$n_1 = 100$, $n_2 = 100$

$\hat{P}_c = ?$

We know that

$$\hat{P}_c = \frac{n_1\hat{P}_1 + n_2\hat{P}_2}{n_1 + n_2}$$

$$= \frac{(100)(0.31) + (100)(0.24)}{100 + 100}$$

$$= \frac{31 + 24}{200} = \frac{55}{200}$$

$$\hat{P}_c = 0.275$$

Q No2 If $E(X) = 0.25$; then find $E(2X)$

We are given that

$E(X) = 0.25$

We have to find

$E(2X) = ?$

$$E(2X) = 2E(X)$$
$$= 2(0.25)$$
$$E(2X) = 0.5$$

Q No 3 Given $N=10$, $n=4$ & $K=5$
find $E(X)$, $Var(X)$?

we are given that
$N=10$, $n=4$, $K=5$

&
we have to find
$E(X) = ?$, $Var(X) = ?$
we know that, the given data is in hypergeometric distribution, So

$$E(X) = \frac{nK}{N}$$

$$= \frac{(4)(5)}{10} = \frac{20}{10} = 2$$

$$\boxed{E(X) = 2}$$

$$Var(X) = \frac{nK}{N}\left(1 - \frac{K}{N}\right)\left(\frac{N-n}{N-1}\right)$$

$$= \frac{(4)(5)}{10}\left(1 - \frac{5}{10}\right)\left(\frac{10-4}{10-1}\right)$$

$$= \frac{\cancel{20}}{\cancel{10}}\left(\frac{5}{10}\right)\left(\frac{6}{9}\right)$$

$$= \frac{6}{9}$$

$$\boxed{Var(X) = 0.6667}$$

QNo4 Name the measures of dispersion that are not based on all values?

1- Range
2- Quartile deviation

QNo5 What is a poisson distribution?

Poisson distribution is also known as rare events distribution. Poisson distribution is limiting form of binomial distribution under conditions:

* 'n' trials are efficiently large
* 'p' very small → 0
* $0.05 \leq np \leq 20$

The probability function of poisson distribution is

$$P(X=x) = \frac{e^{-\mu}\mu^{x}}{x!}$$

$$x = 0, 1, 2, \ldots\ldots \infty$$

Q.No6 Write range, parametre and Formulas of hypergeometric distribution.

Range

Its range is $0, 1, 2, \ldots\ldots n \leq N$

Parametres

It have three parametres

$$n, k, N$$

Formulas

The distribution function is

$$P(X=x) = \frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$$

mean $= \frac{nk}{N}$

variance $= \frac{nk}{N}\left(1-\frac{k}{N}\right)\left(\frac{N-n}{N-1}\right)$

Q#07 Write Range, parameters and formulas for binomial distribution

Range

It range is from $0, 1, 2, \ldots n$

Parameters

It have only 2 parameters

n, p

Formulas

The distribution function is

$$P(X=x) = \binom{n}{x} p^x q^{n-x}$$

mean $= np$

variance $= npq$

Prob 8 A continuous random variable X that can assume between $x=1$ and $x=4$ has a density function given by

$$f(x) = \frac{1}{3}$$

a) Show that the area under curve is equal to 1

b) Find $P(1.5 < X < 3)$

a) we have to show that

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

So

$$\int_{1}^{4} \frac{1}{3}\,dx = 1 \quad , \quad \frac{1}{3}\int_{1}^{4} dx = 1$$

$$\frac{1}{3}[x]_1^4 = 1$$

$$\frac{1}{3}[4-1] = 1$$

$$\frac{1}{3}(3) = 1$$

$$1 = 1$$

proved

b) $P(1.5 < X < 3)$

$$P(1.5 < X < 3) = \int_{1.5}^{3} \frac{1}{3} dx$$

$$= \frac{1}{3} \int_{1.5}^{3} dx$$

$$= \frac{1}{3} [x]_{1.5}^{3}$$

$$= \frac{1}{3} [3 - 1.5] = \frac{1}{3} [1.5]$$

$$= 0.5$$

Q#09 Critical region for following hypothesis

$H_0 : \sigma^2 = 25$

$H_1 : \sigma^2 \neq 25$

where $\alpha = 0.05$, $n = 25$

Critical region

$|z| < z_{1-\alpha/2}$

~~t-test~~ $\alpha = 0.05$, $\alpha/2 = 0.025$

$$1-\frac{\alpha}{2} = 0.975$$

$$z_{1-\alpha/2} = 1.96$$

$$|z| < 1.96$$

Q No 10 If $E(X) = 10$, $E(Y) = 9$
find $E\left(\frac{5}{10}X + \frac{2}{6}Y\right)$

We are given that
$E(X) = 10$, $E(Y) = 9$
We have to find
$= E\left(\frac{5}{10}X + \frac{2}{6}Y\right)$

$$= \frac{5}{10}E(X) + \frac{2}{6}E(Y)$$

$$= \frac{5}{10}(10) + \frac{2}{6}(9)$$

$$= 5 + 3 = 8$$

Q#11 In a normal distribution data How much that fall in interval $\mu \pm 3\sigma$

In normal distribution 99.73% dat fall in the $\mu \pm 3\sigma$ interval

اور $\mu \pm \sigma$ یہاں پہ $\mu \pm 2\sigma$ بھی لکھ سکتے ہیں۔

$\mu \pm \sigma = 68.27\%$

$\mu \pm 2\sigma = 95.45\%$

Q#12 A random sample of size 12 is drawn from normal population with mean 10.9 and variance σ^2 If $s=8$, $\bar{x}=6.4$, using information complete t-statistic.

≠ we are given that

$n=12$, $\mu_0=10.9$, $\hat{s}=8$, $\bar{x}=6.4$

we know that

$$t = \frac{\bar{x} - \mu_0}{\hat{s}/\sqrt{n}}$$

$$= \frac{6.4 - 10.9}{8/\sqrt{12}}$$

$$= \frac{-4.5}{2.309} = -1.948$$

Q#013 Assume That X has a poisson distribution with variance 2
Calculate $P(X=2)$

we know That
The distribution function of poisson distribution is

$$P(X=x) = \frac{e^{-\mu} \mu^x}{x!}$$

We are given That
Variance = 2
In poisson distribution
Variance = mean = $\mu = 2$

So

$$P(X=2) = \frac{e^{-2}(2^2)}{2!}$$

$$= \frac{(0.1353)(4)}{2}$$

$$= 0.2706$$

Q No 14

Let T_1 and T_2 are Two unbiased estimators. variance is

$V(T_1) = \frac{11\sigma^2}{9}$ and $Var(T_2) = \frac{13\sigma^2}{9}$

which estimator is more efficient and why?

~~We know~~

we are given that

$Var(T_1) = \frac{11\sigma^2}{9}$, $Var(T_2) = \frac{13\sigma^2}{9}$

we know that

$$E_f = \frac{Var(T_2)}{Var(T_1)} = \frac{13\sigma^2/9}{11\sigma^2/9}$$

$$= \frac{13\sigma^2}{9} \times \frac{9}{11\sigma^2}$$

$$= \frac{13}{11} \text{ which is}$$

greater than 1, showing that $Var(T_1) < Var(T_2)$. Thus T_1 is more efficient than T_2

Q No 14

Find value of χ^2 test statistic if $n = 2.5$, $s^2 = 3$, $\sigma_0^2 = 18$

we know that

$$\chi^2 = \frac{ns^2}{\sigma^2}$$

$$= \frac{(2.5)(3)}{18} = 0.4167$$

Q No 16 Write variance of hypergeometric distribution.

$$\text{Variance} = \frac{nk}{N}\left(1 - \frac{k}{N}\right)\left(\frac{N-n}{N-1}\right).$$

Q No17 Find proportion of X_1 & X_2 if

$X_1 = 300$, $n_1 = 400$

$X_2 = 200$, $300 = n_2$

Proportion of X_1

$$P_1 = \frac{X_1}{n_1} = \frac{300}{400} = \frac{3}{4} = 0.75 = 75\%$$

Proportion of X_2

$$P_2 = \frac{X_2}{n_2} = \frac{200}{300} = \frac{2}{3} = 0.667 = 66.7\%$$

Q No18 What is Joint distribution?

The distribution of Two or more random variables which are observed simultanously when an experiment is performed, is called Joint distribution.

Q No19 $n = 1150$, $x = 450$, $P_0 = 0.39$

$H_0 : P_0 = 0.39$, $\alpha = 0.01$

Stated hypothesis.

1- Formulation

$H_0 : P_0 = 0.39$

$H_1 : P_0 \neq 0.39$

2- LOS

$\alpha = 0.01$, $\alpha/2 = 0.01/2 = 0.005$, $1 - \frac{\alpha}{2} = 0.995$

3- Test statistic

$$z = \frac{X - np_0}{\sqrt{np_0 q_0}} \quad , \quad z = \frac{(X \pm 1/2) - np_0}{\sqrt{(np_0 q_0)}}$$

4- Critical region

$$z_{1-\alpha/2} = z_{0.995} = 2.576$$

$$|z|_c < 2.576$$

5- Calculation

we are given that

$n = 1150$, $X = 450$, $p_0 = 0.39$

$q_0 = 1 - p_0 = 1 - 0.39 = 0.61$

~~$z = 4.50$~~

$$z = \frac{(450 - 1/2) - (1150)(0.39)}{\sqrt{(1150)(0.39)(0.61)}}$$

$$= \frac{449.5 - 448.5}{16.540}$$

$$= \frac{1}{16.540} = 0.0604$$

$$z = 0.0604$$

b) **Conclusion**

Since $|z|_c < 2.576$

$0.0604 < 2.576$

It does not lies in rejection region so we do not reject H_0 against H_1

Note:

ہم z لکھے Formula ہے Test statistic ایک سی $\frac{1}{2}$ ہے ایک سی ہیں - جس سی ہے "کیے" continuity correction اسے اور یہ وہاں ہوگا use اگر ہمارے پاس

$$X > np_0$$

تو question سی X جو ہے وہ 450 اور np_0 آیا ہے 448.5 تو X بڑا ہوا اس سے - سی X ہے سو -1/ سی calculation

QNO20 what is relationship b/w level of significance and critical region?

A critical region specifies a set of values of the test statistic for which null hypothesis is rejected.

The actual type I error that we get by using critical region it will be the level of significance-

QNO21 what is g(x) and h(y)?

From the joint probability function for (X,Y), we can obtain the individual probability function of X and Y.

∴ individual probability function ka matlb hai X is alg or y ke alg probability

Such individual probability function are called marginal probability function.

The marginal p.f of X

$g(x) = P(X=x)$

The marginal p.f of Y

$h(y) = P(Y=y)$

Qno22 Define point estimation-

The object of point estimation is To obtain a single number from the sample of unknown true value of population parameter.

Qno23 Write 3 properties of normal distribution.

1- **Continuous distribution:**

The normal probability distribution is a continuous distribution

That ranges from $-\infty$ To $+\infty$

$$R_x = \{ x : -\infty < x < +\infty \}$$

2- **Total probability**

The total area under the cure is unity. That is

$$P(-\infty < X < +\infty) = 1$$

3- **Symmetrical distribution:**

The normal distribution is symmetrical.

* mean = median = mode
* $x_{0.75} - \mu = \mu - x_{0.25}$
* $\mu_1 = \mu_3 = \mu_5 = \cdots = 0$

Q_1 اور Q_3، $x_{0.25}$ اور $x_{0.75}$ بھی normal distribution کے لئے symmetrical ہیں۔

Q No 24 Why we use pooled proportion instead of individual p?

We use the pooled proportion instead of individual proportion, because the pooled proportion estimate the standard error and individual not-

Q No 15 What is difference b/w random design and random design block (random block design).

Completely randomized design-

A completely randomized design (CR), which is the simplest type of the basic designs, may be defined as a design in which treatments are assigned to experimental units completely at random, that is the randomization is done without any restrictions.

Randomized complete block design.

A randomized complete block (RCB) design may be defined as on in which.

i) The experimental metrial is divided into groups or block.

ii) Each block contains a complete

set of treatments

iii) The treatments are assigned at random to the experimental units within restrictions.

Q No26 Can we control type I error and type II error?

To control Type I error we can set simple a lower significance level.

While Type II error is immposible to control completely, The only possible chance to control Tye II error by increasing sample size.

Q No27 $n = 15$, $\alpha = 95\%$ Then find $t_{\alpha/2;\nu} = ?$

we are given that

$n = 15$, $\alpha = 95\% = 0.15$

$\alpha/2 = 0.95/2 = 0.075$

$\nu = n - 1 = 15 - 1 = 14$

$t_{0.025;14} = 2.145$

Ans 28 If $TSS = 272.25$

$\sum_i \sum_j x^2 = 4953$

Find (C.F) ~~Coefficient~~ correction factor -

we know that

$$TSS = \sum_i \sum_j x^2_{ij} - C.F$$

$272.25 = 4953 - C.F$

$C.F = 4953 - 272.25$

$C.F = 4680.75$

Ans 29 what is relation b/w mean and variance in poisson distributio

The mean and variance in the poisson distribution are same i.e. μ.

Q No30 The department claims that the exceeds Rs. 2500 at the 0.05 level of, then formulate null alternative hypothesis?

Formulation:

$H_0 : \mu < 2500$

$H_1 : \mu > 2500$

Q No31 When expected frequency in any category is less then 5 what we use?

If an expected frequency in any category is less then 5, then we use Yate' correction of continuity

$$\chi^2 = \sum_{i=1}^{2} \frac{(|o-e| - \frac{1}{2})^2}{e}$$

it is applied ~~in~~ e

it is applied in the normal approximation to the binomial distribution.

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Q.No 32 ~~If $\bar{x} = 800$,~~

If $n = 400$, $x = 275$, then find 95% confidence interval for population proportion.

we know that

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

we are given that

$$n = 400, \quad x = 275$$

$$\hat{p} = \frac{x}{n} = \frac{275}{400} = 0.69$$

$$\hat{q} = 1 - \hat{p} = 1 - 0.69 = 0.31$$

$$1 - \alpha = 0.95 \Rightarrow \alpha = 0.05, \quad \alpha/2 = 0.025$$

$$0.69 \pm z_{(0.025)} \sqrt{\frac{(0.69)(0.31)}{400}}$$

$$0.69 \pm z_{0.025} (0.023)$$

$$0.69 \pm (1.96)(0.023)$$

$$0.69 \pm 0.04508$$

$$0.69 + 0.04508 \quad , \quad 0.69 - 0.04508$$

$$0.73508 \quad , \quad 0.645$$

(0.645, 0.73508)

Q No33

Explain Chi-Square method?

The chi-square distribution contains only one parameter, called the number of degree of freedom (df), where the term d.f represents the number of independent random variable that expresses the chi-square.

$$\chi^2 = \sum_{i=1}^{2} \frac{(|O-e| - e)^2}{e}$$

Q No34 Why the statistics is called science of decision making. Explain it?

[redacted]

[redacted]

9 n This it is called ~~a~~ science of decision making because it is enabling us to establish numerical benchmarks and monitor and evaluate The progress of our policy or program.

Best of Luck

11-9-2021